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Normative values for the Measure of strength and Body composition in Athletes: A cross-sectional Study

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ABSTRACT

Background: Handgrip strength and explosive strength are key indicators of athletic performance, yet no normative data exist for trained Lebanese athletes. Establishing reference values for these strength parameters is essential for talent identification, performance evaluation, and training optimization. **Objective:** This study aimed to establish normative reference values for handgrip strength, explosive strength, and body composition in trained and untrained Lebanese males and females. **Methods:** This cross-sectional study included 264 participants (162 males, 102 females) aged 18–40, categorized as trained (>300 min/week of resistance training) or untrained (<300 min/week). Body composition metrics—including skeletal muscle mass (SMM), fat-free mass (FFM), and body fat percentage (PBF)—and upper-limb handgrip performance variables (maximal grip strength [Fmax] and rate of force development [RFDmax]) were assessed using a standardized handgrip protocol. Statistical analyses included Multivariate Analysis of Variance (MANOVA) to compare groups, and a seven-class normative system was used for performance classification. **Results:** Significant differences ($p \leq 0.05$) were observed between trained and untrained individuals, particularly among females. Trained females exhibited a 17.04% increase in relative RFD (RFDrel) and a 12.98% improvement in maximal force (Fmax) compared to untrained females, along with higher SMM and lower PBF. Trained males demonstrated more modest improvements, with Frel_SUM increasing by 5.53%. Normative reference values were established for handgrip strength and explosive strength, categorizing performance levels from "Very Poor" to "Superior." **Conclusion:** This study provides the first normative values for handgrip strength, explosive strength, and body composition in Lebanese athletes. These benchmarks will aid in athlete evaluation, talent identification, and training program development. Further research should explore longitudinal training adaptations and the effects of specific training modalities on strength performance.

Key words: Body Composition, Hand Strength, Muscle Strength, Reference Values, Athletes, Lebanon

INTRODUCTION

Strength and explosive strength are critical for athletic performance, particularly in sports requiring high-intensity movements such as volleyball, basketball, handball, and weightlifting. These physical attributes significantly impact an athlete's ability to jump, sprint, and change direction, all of which are crucial for competitive success (Suchomel et al., 2016; Comfort et al., 2018). Despite the well-established importance of strength and explosiveness, limited research has explored how these attributes vary across trained and untrained individuals, especially within specific populations. Understanding the impact of resistance training on muscle strength, explosive power, and body composition is essential for optimizing athletic performance. Additionally, sex-based differences in training adaptations remain unclear, as hormonal and physiological differences may influence responses to resistance training (Haun et al., 2019).

A key limitation in the field is the lack of established reference values for handgrip strength and explosive strength, particularly in Lebanese athletes. Establishing such normative data is crucial for talent identification, performance evaluation, and training optimization. Without these benchmarks, it becomes difficult to assess whether an athlete meets competitive standards or requires targeted training interventions (Zatsiorsky & Kraemer, 2006). Given these gaps in the literature, this study aims to investigate differences in body composition, handgrip strength, and explosive strength between trained and untrained Lebanese males and females and to establish normative reference values for handgrip and explosive strength. These findings will contribute to sports science by providing a benchmark for athlete evaluation and training program development.

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Assessing strength and explosive strength is essential for both sports performance and general health. Previous studies have demonstrated the significance of handgrip strength as a predictor of overall muscular strength, neuromuscular performance, and health status (Dodds et al., 2016; Buckner et al., 2017). Furthermore, handgrip strength is particularly relevant for sports requiring upper-body strength, grip endurance, and ball control, such as volleyball, basketball, and handball (Chaouachi et al., 2009). Despite its recognized importance, there are no established reference values for handgrip strength and explosive strength in Lebanese athletes. Establishing these benchmarks is necessary for coaches, sports scientists, and strength and conditioning professionals to evaluate athletes' performance objectively. Such data can facilitate talent identification, enable performance comparisons across populations, and inform individualized training strategies to enhance athletic capabilities (Zatsiorsky, 1982). Moreover, understanding the impact of resistance training on body composition and strength is essential for optimizing training programs. Resistance training has been shown to increase skeletal muscle mass (SMM) and decrease body fat mass (BFM), but these adaptations may differ based on training history and sex (Ihalainen et al., 2019). Investigating these differences will help refine training protocols to maximize strength gains and minimize injury risk.

Prior research has extensively examined the role of resistance training in improving strength and body composition. Studies have shown that consistent resistance training leads to neuromuscular adaptations, enhancing maximal force production and rate of force development (Grgic et al., 2018). However, these adaptations are influenced by factors such as training frequency, intensity, and sex-related physiological differences (West et al., 2012). Handgrip strength has been widely used as an indicator of overall muscular function and athletic performance. Previous studies have demonstrated that grip strength correlates with explosive power and can be used as a practical assessment tool for evaluating strength capabilities (Cronin et al., 2007). Additionally, it has been shown that trained individuals typically exhibit higher grip strength and explosive strength than their untrained counterparts, highlighting the impact of resistance training (Andraos & El Mdawar, 2024). Research suggests that males generally experience greater fat reduction, whereas females show more pronounced increases in lean muscle mass following resistance training (Huang et al., 2020). These findings suggest that training programs should be tailored to account for sex-specific physiological responses to optimize performance outcomes.

Despite growing interest in strength assessment, there is currently no normative data specific to the Lebanese population for key performance measures such as handgrip strength and explosive strength, which are widely recognized indicators of muscular fitness, health status, and athletic potential. Existing reference values are mainly derived from Western or Asian populations, whose physiological characteristics and training contexts may differ significantly from those of Lebanese individuals. This lack of localized data limits the ability of coaches, sports scientists, and health professionals to accurately evaluate Lebanese athletes, identify talent, and design individualized training or rehabilitation programs. Therefore, the present study aims to establish population-specific reference values for handgrip strength and explosive strength among Lebanese adults, comparing trained and untrained individuals and examining sex-based differences in performance and body composition.

To test these hypotheses, a cross-sectional study design was implemented, comparing body composition and strength parameters between trained and untrained Lebanese athletes. Handgrip strength and explosive strength were assessed using standardized protocols to ensure reliability and accuracy. The study also aimed to establish normative values for strength performance, categorizing athletes into performance levels ranging from "Very Poor" to "Superior" based on their results. By addressing these objectives, this research provides valuable insights into the impact of resistance training on strength development, contributes to the scientific understanding of sex-based training adaptations, and establishes normative benchmarks for Lebanese athletes. These findings will help coaches, trainers, and sports professionals develop more effective, individualized training programs tailored to athletes' specific needs.

METHOD

Participant and Study Design

This study employed a cross-sectional observational design, conducted at Antonine University, Baabda, Lebanon, in the Laboratory of the 3S: Sport, Santé, Société (L3S). A total of 369 individuals (242 males and 127 females) were initially recruited from various regions across Lebanon. All participants completed a pre-test questionnaire collecting demographic data, training history, weekly training frequency, session duration, and type of physical activity (e.g., resistance training, aerobic training). The research complied with the principles of the Declaration of Helsinki and was approved by the Ethics Committee of the University of Belgrade, Faculty of Sport and Physical Education (approval number: 02 No. 484-2).

No a priori sample size calculation was performed due to the absence of prior normative data on handgrip strength and body composition in the Lebanese population. Instead, recruitment aimed to obtain a sufficiently large convenience sample to establish initial reference values and enable sex- and training-status comparisons. Future studies should conduct power analyses using the variability and effect sizes observed in the present data to refine sample size requirements for hypothesis testing.

Participants were required to be between 18 and 40 years old and free from any injuries or medical conditions that could affect their strength and body composition. The age range was limited to 18–40 years because this study aimed to establish reference values relevant for competitive athletes, as individuals over 40 rarely compete at elite levels. However, participants over 40 years old were initially included during recruitment to allow for potential future research on strength and body composition changes in older

populations. Individuals with hormonal disorders, injuries, or other health conditions that could interfere with strength measurements were excluded. After applying these exclusion criteria, 105 individuals were removed from the study, resulting in a final sample of 264 participants, consisting of 162 males (79 untrained, 83 trained) and 102 females (32 trained, 74 untrained).

Participants were classified as trained or untrained according to their resistance training background, taking into account both weekly volume and previous experience. Individuals were considered *trained* if they reported performing structured resistance training for more than 300 minutes per week, had maintained at least one year of consistent practice, and followed moderate- to high-intensity protocols (≥60% of 1-RM) that targeted the major muscle groups. This group included recreational lifters as well as athletes involved in sports where resistance training forms a systematic component of preparation (such as weightlifting, football, or combat sports). By contrast, the *untrained* group comprised participants who reported less than 300 minutes of resistance training per week and had no sustained training history over the preceding 12 months, typically engaging only in occasional or non-structured activity. To ensure clarity in the classification, individuals who reported exactly 300 minutes of training per week were excluded from the analysis.

All procedures were approved by the Antonine University Ethical Committee (Code: 413, Reference Number: 1675-2023) and were conducted in accordance with the Declaration of Helsinki (Christie, 2000).

Basic characteristics of the sample are shown in Table 1.

Sampling Procedures

Participants were recruited from various regions across Lebanon to ensure a diverse representation of trained and untrained individuals. A stratified sampling approach was used, where participants were first categorized based on their training status (trained vs. untrained) and then recruited voluntarily.

Recruitment was conducted through sports clubs, gyms, universities, corporate offices, and banks, ensuring the inclusion of individuals with varied training backgrounds and occupational settings. No financial compensation was provided for participation. All participants provided informed consent before taking part in the study.

Table 1. Descriptive statistics of anthropometric characteristics in the lebanese population (aged 18-40) by gender

Variable	Male (Mean±SD) *n*=162	Female (Mean±SD) *n*=106
Age (year)	24.83±6.40	24.69±6.17
BH (cm)	177.95±7.70	163.93±6.36
BM (kg)	81.02±14.98	62.67±11.30
BMI (kg/m^2)	25.57±3.95	23.31±4.01

Sample size, power, and precision

No a priori sample size calculation was performed for this study. Recruitment aimed to collect a sufficiently large convenience sample to establish initial normative reference values for Lebanese athletes. Given the exploratory nature of the research and the absence of prior local data on handgrip strength and body composition in this population, it was not feasible to calculate an expected effect size in advance. Future studies should perform power analyses based on the variability and effect sizes observed in the present data to confirm appropriate sample sizes for hypothesis testing.

Measures and covariates

Anthropometric and strength assessments were conducted under controlled conditions at the Laboratory of the 3S: Sport, Santé, Société (L3S). All measurements were performed by a qualified nutritionist following standardized protocols. The anthropometric variables included body mass (BM, kg), body height (BH, cm), body mass index (BMI, kg/m²), body fat mass (BFM, kg), fat-free mass (FFM, kg), skeletal muscle mass (SMM, kg), percentage of body fat (PBF, %), and percentage of skeletal muscle mass (PSMM, %). Body composition was assessed using bioelectrical impedance analysis (BIA, i30, Mediana, Korea), which provides estimates of fat mass and muscle mass (Rakic et al., 2022). Height was measured using a portable digital height scale (Campry, China).

Upper-limb handgrip strength was measured using a standardized handgrip dynamometer (Sports Medical Solution, Belgrade, Serbia), connected to a laptop-based software–hardware system for isometric testing (Sports Medical Solutions Isometrics, ver. 3.4.0). The device automatically recorded both maximal force (Fmax) and rate of force development (RFD), providing direct output values for each trial. Previous research has confirmed the high reliability of this system, with intraclass correlation coefficients ranging from 0.938 to 0.977 for Fmax and 0.903 to 0.971 for RFDmax (Andraos & El Mdawar, 2024). Participants were seated upright with one arm extended in a neutral position and slightly abducted for the hand being tested. Prior to testing, each participant completed two practice trials with both the dominant and non-dominant hand. During the assessment, participants were instructed to apply maximum grip force as quickly and forcefully as possible, sustaining the contraction for at least two seconds after the signal. Standardized verbal encouragement was provided throughout to ensure maximal effort. Each hand was tested twice in a randomized order, with a one-minute rest period between trials.The primary strength-related outcomes included maximal force (Fmax, N), defined as the peak force output; rate of force development (RFDmax, N/s), representing the maximum slope of the force–time curve and automatically computed by the dynamometer software; relative force (Frel, N/kg), calculated by normalizing maximal force to body weight; and relative rate of force development (RFDrel, N/s/kg), derived by normalizing RFD to body weight. By using a dynamometer with integrated software, Strength & explosive strength metrics (F_{max}, F_{rel}, RFD_{max} and RFD_{rel}) were calculated automatically

from the force–time curve, ensuring standardized and reproducible measurement across participants.

Research Design

This study followed a cross-sectional research design, comparing body composition and strength parameters between trained and untrained Lebanese males and females. Participants were not assigned to experimental conditions, and no interventions were applied. Instead, existing training behaviors were analyzed to examine their impact on handgrip strength and explosive strength.

Statistical Analyses

Descriptive statistics, including mean (M) and standard deviation (SD), were calculated for all variables. A Multivariate Analysis of Variance (MANOVA) was conducted to compare trained and untrained individuals across body composition characteristics and handgrip strength parameters. Key comparisons focused on maximal force (Fmax) and rate of force development (RFDmax) in both the dominant and non-dominant hands, as well as the summed values for both hands (Fmax_SUM and RFDmax_SUM). In addition, both absolute and relative strength measures (Frel and RFDrel) were analyzed. To establish normative values, a seven-class classification system was applied, categorizing performance levels as Very Poor, Poor, Below Standards, Standards, Above Standards, Excellent, and Superior, in accordance with established metrological procedures (Zatsiorsky, 1982).

Statistical significance was defined as $p < .05$ (two-tailed). P-values between.05 and.10 were reported as trends, indicating a possible tendency, but were not considered statistically significant. Because multiple dependent variables were analyzed, the increased risk of Type I error is acknowledged; however, no formal correction (e.g., Bonferroni) was applied, and univariate follow-up results are therefore interpreted cautiously, with emphasis placed on the overall multivariate effects.

All statistical analyses were performed using SPSS Win Statistics 26.0 and Excel 2016.

RESULTS

Table 2 presents the mean values and standard deviations of anthropometric and body composition variables for trained and untrained males and females within the Lebanese population. A multivariate analysis of variance (MANOVA) revealed a significant overall effect of training status on the combined dependent variables for both males, *Wilks' Λ* = .418, $F(13, 148) = 15.864$, $p < .001$, and females, *Wilks' Λ* = .314, $F(13, 92) = 15.485$, $p < .001$, indicating substantial multivariate differences in body composition profiles between trained and untrained individuals.

In males, univariate analyses indicated that trained participants exhibited significantly lower fat mass index, $F(1, 160) = 7.616$, $p = .006$, and body fat mass, $F(1, 160) = 6.766$, $p = .010$, as well as significantly higher skeletal muscle mass, $F(1, 160) = 4.013$, $p = .047$, percent skeletal muscle mass, $F(1, 160) = 9.193$, $p = .003$, skeletal muscle mass index, $F(1, 160) = 4.289$, $p = .040$, and body composition index, $F(1, 160) = 6.056$, $p = .015$. Additionally, trained males showed significantly lower percent body fat, $F(1, 160) = 8.759$, $p = .004$.

Among females, trained individuals demonstrated significantly greater fat-free mass, $F(1, 104) = 5.858$, $p = .017$, skeletal muscle mass, $F(1, 104) = 22.035$, $p < .001$, percent skeletal muscle mass, $F(1, 104) = 37.400$, $p < .001$, skeletal muscle mass index, $F(1, 104) = 23.937$, $p < .001$, body composition index, $F(1, 104) = 28.758$, $p < .001$, and muscle-fat index, $F(1, 104) = 25.598$, $p < .001$. Percent body fat was also significantly lower in trained females, $F(1, 104) = 10.500$, $p = .002$.

Finally, average weekly training duration was significantly higher in trained versus untrained individuals for both males, $F(1, 160) = 183.359$, $p < .001$, and females, $F(1, 104) = 183.601$, $p < .001$, further validating the training status classification.

Table 3 summarizes the handgrip strength comparison between trained and untrained males. The multivariate analysis was not significant, $F(8, 153) = 1.807$, $p = .080$. Univariate tests showed that trained males had higher non-dominant maximal force, $F(1, 160) = 5.124$, $p = .025$, non-dominant relative force, $F(1, 160) = 6.162$, $p = .014$, and Fmax_SUM, $F(1, 160) = 3.947$, $p = .049$.

Table 4 presents results for females, with a significant multivariate effect, $F(8, 97) = 3.196$, $p = .003$. Trained females outperformed untrained females in all variables (e.g., Fmax_SUM, $F(1, 104) = 12.263$, $p = .001$; RFDmax_SUM, $F(1, 104) = 13.387$, $p < .001$; RFDrel_SUM, $F(1, 104) = 17.110$, $p < .001$).

Table 5 establishes normative values for handgrip strength and explosive strength in trained Lebanese athletes, categorized by performance levels from very poor to supe-rior. These values provide a benchmark for evaluating grip strength and explosive power in strength-based sports.

DISCUSSION

The primary objective of this study was to establish normative reference values for body composition, handgrip strength, and handgrip explosive strength among Lebanese adults. The results demonstrated that trained individuals outperformed untrained ones across most parameters, with clear sex-related differences in both body composition and neuromuscular performance. These findings are consistent with previous research emphasizing the impact of resistance training on morphological adaptations and strength development (Grgic et al., 2018; Haun et al., 2019).

In terms of body composition, trained males and females exhibited significantly higher skeletal muscle mass (SMM), fat-free mass (FFM), and protein-related SMM (PSMM), along with lower percentages of body fat (PBF). The relative increases in SMM and PSMM were particularly notable in females (15% and 14.9%, respectively), compared to males (6.7% and 7.7%). This suggests that women may experience greater proportional adaptations in response to training, a finding supported by Haun et al. (2019), who reported that

Table 2. Multivariate comparison of anthropometric and body composition variables between trained and untrained males and females in the lebanese population

Variable	Untrained Male (Mean±SD) *n*=79	Trained Male (Mean±SD) *n*=83	*p-Value* (Male)	Untrained Female (Mean±SD) *n*=74	Trained Female (Mean±SD) *n*=32	*p*-Value (Female)
BH (cm)	177.52±6.93	178.35±8.39	0.490	163.65±6.13	164.69±6.90	0.445
BM (kg)	81.50±16.94	80.56±12.93	0.691	62.35±11.02	63.43±12.08	0.652
BMI (kg/m²)	25.83±4.65	25.31±3.15	0.405	23.31±4.12	23.34±3.85	0.965
FFM (kg)	63.46±9.51	65.96±9.24	0.093	41.32±6.84	44.73±6.18	0.017
FMI (kg/m²)	5.71±3.00	4.59±2.09	0.006	8.07±2.99	6.91±3.08	0.075
BFM (kg)	18.03±9.82	14.60±6.76	0.010	21.46±7.54	18.70±8.62	0.101
SMM (kg)	35.43±6.12	37.33±5.92	0.047	19.32±4.29	23.65±4.50	0.000
PSMM (%)	44.09±5.74	46.53±4.46	0.003	31.07±4.89	37.52±5.19	0.000
PBF (%)	21.01±8.01	17.67±6.31	0.004	33.60±7.09	28.71±7.23	0.002
SMMI (kg/m²)	11.21±1.61	11.71±1.48	0.040	7.20±1.49	8.67±1.24	0.000
IBC (AU)	1.39±0.52	1.61±0.58	0.015	0.70±0.085	0.84±0.19	0.00
MFI	2.67±1.71	3.14±1.63	0.077	0.97±0.29	1.44±0.66	0.000
AWTD (min)	102.06±105.33	579.30±295.89	0.000	69.32±93.68	572.89±287.82	0.000

BH= Body Height; BM= Body Mass; BMI= Body Mass Index; FFM= Fat-Free Mass; FMI= Fat Mass Index; BFM= Body Fat Mass; SMM= Skeletal Muscle Mass; PSMM= Percent Skeletal Muscle Mass; PBF= Percent Body Fat; SMMI= Skeletal Muscle Mass Index; IBC= Index of Body Composition; MFI= Muscle-Fat Index; AWTD= Average Weekly Training Duration

Table 3. Comparison of handgrip strength and explosive strength variables between trained and untrained males

Variable	Untrained (Mean±SD) *n*=79	Trained (Mean±SD) *n*=83	Percentage Difference Between Groups (%)	*p*-Value
$DHGF_{max}$ (N)	487.51±89.82	508.92±85.33	4.2%	0.122
NDHG F_{max} (N)	454.32±88.41	486.42±91.93	6.6%	0.025
$DHGF_{rel}$ (N/kg)	6.10±1.07	6.38±0.97	4.38%	0.079
NDHG $_{Frel}$ (N/kg)	5.68±1.06	6.09±0.99	6.73%	0.014
DHG RFD_{max} (N/s)	3051.35±712.52	3056.69±618.26	0.17%	0.959
NDHG RFD_{max} (N/s)	2801.96±750.30	2949.29±641.09	5.0%	0.180
DHG RFD_{rel} (N/kg/s)	38.01±7.93	38.48±7.95	1.22%	0.706
NDHG RFD_{rel} (N/kg/s)	34.85±8.16	37.08±7.93	6.01%	0.080
F_{max}—SUM (N)	941.82±171.82	995.34±170.94	5.37%	0.049
RFD_{max}—SUM (N/s)	5853.32±1408.99	6005.98±1188.10	2.54%	0.456
F_{rel}—SUM (N/kg)	11.78±2.05	12.47±1.88	5.53%	0.028
RFD_{rel}—SUM (N/kg/s)	72.86±15.28	75.56±14.97	3.57%	0.258

women often display more pronounced relative gains in lean mass, especially when starting from lower baseline values.

With respect to strength parameters, trained males displayed significantly higher maximal force (Fmax) in the dominant hand, whereas no significant differences were observed in explosive strength (RFD). Conversely, trained females demonstrated significant improvements in both maximal and explosive handgrip force compared to untrained females. These sex-specific adaptations can be explained by underlying physiological mechanisms. Early training responses in women are often characterized by pronounced neural adaptations—including enhanced motor unit recruitment, synchronization, and firing frequency—that contribute to substantial gains in both maximal and explosive force (Maffiuletti et al., 2016). Furthermore, given that women typically begin with lower absolute strength and muscle mass, their relative percentage improvements following resistance training tend to be greater than those observed in men (Ihalainen et al., 2019; West et al., 2012).

From a practical perspective, these findings indicate that women can achieve robust improvements in strength and power even through standard resistance training programs. In contrast, the limited gains in RFD observed among trained males suggest that traditional hypertrophy- or general strength–focused regimens may be insufficient for optimizing explosive force development. Since RFD is best enhanced by high-velocity and power-specific training modalities—such as plyometric and ballistic resistance exercises—these should be deliberately integrated into male training programs to maximize adaptations (Jaric, 2002;

Table 4. Comparison of handgrip strength and explosive strength variables between trained and untrained females

Variable	Untrained (Mean±SD) *n*=74	Trained (Mean±SD) *n*=32	Percentage Difference Between Groups (%)	*p-Value*
DHG F_{max} (N)	267.34±55.42	302.41±48.83	11.60%	0.003
NDHG F_{max} (N)	240.57±56.94	281.59±44.05	14.57%	0.000
DHG F_{rel} (N/kg)	4.36±0.93	4.86±0.93	10.29%	0.011
NDHG $_{Frel}$ (N/kg)	3.91±0.86	4.55±0.93	14.07%	0.001
DHG RFD_{max} (N/s)	1515.50±457.85	1833.13±384.95	17.30%	0.001
NDHG RFD_{max} (N/s)	1381.62±451.08	1710.12±371.03	19.20%	0.000
DHG RFD_{rel} (N/kg/s)	24.57±6.89	29.65±7.74	17.12%	0.001
NDHG RFD_{rel} (N/kg/s)	22.35±6.61	27.74±7.35	19.44%	0.000
F_{max}–SUM (N)	507.91±108.04	584.00±88.89	12.98%	0.001
RFD_{max}–SUM (N/s)	2897.12±878.70	3543.25±720.42	18.24%	0.000
F_{rel}–SUM (N/kg)	8.26±1.71	9.41±1.81	12.22%	0.002
RFD_{rel}–SUM (N/kg/s)	46.92±12.84	57.39±14.7	18.24%	0.000

Table 5. Normative values for hand grip strength and explosive strength among trained lebanese male and female athletes

Variables	Very Poor	Poor	Below average	Average	Above average	Excellent	Superior
Males							
DHG F_{max} (N)	≤337	338-423	424-465	466-552	553-594	595-680	≥681
NDHG F_{max} (N)	≤302	303-393	394-439	440-532	533-578	579-670	≥671
DHG F_{rel} (N/kg)	≤4.43	5.40-4.44	5.41-5.89	5.90-6.87	6.88-7.35	7.36 – 8.32	≥8.33
NDHG F_{rel} (N/kg)	≤4.10	4.11-5.09	5.10-5.59	5.60-6.59	6.6-7.08	7.09 – 8.07	≥8.08
DHG RFD_{max} (N/s)	≤1819	1820-2437	2348-2747	2748-3366	3367-3675	3676-4293	≥4294
NDHG RFD_{max} (N/s)	≤1666	1667-2307	2308-2628	2629-3270	3271-3590	3591-4231	≥4 232
DHG RFD_{rel} (N/s/kg)	≤22.57	22.58-30.52	30.53-34.50	34.51-42.46	42.47-46.43	46.44-54.38	≥54.39
NDH RFD_{rel} (N/s/kg)	≤21.21	21.22-29.14	29.15-33.11	33.12-41.05	41.06-45.01	45.02-52.94	≥52.95
F_{max}–SUM (N)	≤622	623-795	796-882	883-1056	1057-1142	1143-1315	≥1316
RFD_{max}–SUM (N/s)	≤3333	3334-4632	4633-5281	5282-6581	6582-7230	7231-8529	≥8530
F_{rel}–SUM (N/kg)	≤8.14	8.15-10.13	10.14-11.13	11.14-13.13	13.14-14.12	14.13-16.11	≥16.12
RFD_{rel}–SUM (N/s/kg)	≤43.97	43.98-59.10	59.11-66.67	66.68-81.81	81.82-89.37	89.38-104.50	≥104.51
Females							
DHG F_{max} (N)	≤204	205-253	254-277	278-327	328-351	352-400	≥401
NDHG F_{max} (N)	≤192	193-237	238-259	260-304	305-326	327-370	≥371
DHG F_{rel} (N/kg)	≤2.99	3.00-3.92	3.93-4.39	4.40 – 5.33	5.34 – 5.79	5.80 – 6.72	≥6.73
NDHG F_{rel} (N/kg)	≤2.68	2.69-3.61	3.62-4.08	4.09-5.02	5.03-5.48	5.49-6.41	≥6.42
DHG RFD_{max} (N/s)	≤1062	1063-1447	1448-1640	1641-2026	2027-2218	2219-2603	≥2604
NDHG RFD_{max} (N/s)	≤967	968-1338	1339-1524	1525-1896	1897-2081	2082-2452	≥2453
DHG RFD_{rel} (N/s/kg)	≤14.16	14.17-21.90	21.91-25.77	25.78-33.52	33.53-37.39	37.40-45.13	≥45.14
NDH RFD_{rel} (N/s/kg)	≤13.03	13.04-20.38	20.39-24.06	24.07-31.42	31.43-35.09	35.10-42.44	≥42.45
F_{max}–SUM (N)	≤314	315-422	423-476	477-585	586-639	640-747	≥748
RFD_{max}–SUM (N/s)	≤1325	1326-2208	2209-2650	2651-3534	3535-3975	3976-4858	≥4859
F_{rel}–SUM (N/kg)	≤4.98	4.99-6.79	6.80-7.70	7.71– 9.52	9.53 – 10.42	10.43-12.23	≥12.24
RFD_{rel}–SUM (N/s/kg)	≤21.67	21.68-35.87	35.88-42.97	42.98 – 57.18	57.19 – 64.28	64.29-78.48	≥78.49

Fmax=maximal muscle force; RFDmax=maximal explosive muscle force; Frel=relative aspect of maximal muscle force; RFDrel=relative aspect of maximal explosive muscle force; D=dominant hand; ND=non-dominant hand

Knudsen et al., 2019). Thus, tailoring program content to the adaptive potential of each sex is likely to yield optimal neuromuscular outcomes.

A further contribution of this study is the development of a percentile-based classification system for handgrip strength and explosive strength, ranging from "Very Poor" to "Superior." This framework enables practitioners to interpret performance within a population-specific context, thereby supporting individualized training prescriptions, rehabilitation monitoring, and performance evaluation. Nonetheless, this scale should be considered preliminary, and its validity would benefit from cross-validation in other populations and age groups.

Several limitations should be acknowledged. First, the cross-sectional design precludes causal inferences. Second, we did not account for nutritional status, detailed training history, or specific modalities used by the trained participants, which may have influenced the results. Third, the sample, although diverse in regional representation and training status, was limited to individuals aged 18–40 years and recruited on a voluntary basis, which could introduce selection bias.

Future research should address these limitations by employing longitudinal designs to track training-induced adaptations over time and by examining the effects of different training modalities on both maximal and explosive grip strength. Expanding investigations to include adolescents, older adults, and elite athletes would also refine and validate normative classification systems, enhancing their generalizability and practical application.

When placed in an international context, the normative values generated in this study are broadly consistent with those reported in European and Asian cohorts, while also reflecting region-specific differences. For instance, large-scale studies have shown that adults from Northern Europe exhibit higher absolute handgrip strength, largely due to greater body size and higher habitual physical activity (Dodds et al., 2014; Massy-Westropp et al., 2011). In contrast, our values align more closely with Mediterranean and Middle Eastern cohorts, where environmental, nutritional, and cultural factors may shape performance outcomes (Alqahtani et al., 2019). Importantly, our recent findings (Andraos & El Mdawar, 2024) demonstrated that Lebanese athletes display normative ranges similar to international standards but with relatively lower RFD, suggesting that regional training practices may emphasize hypertrophy and general conditioning over high-velocity or ballistic strength development. Such cultural and physiological influences—including differences in body composition, training traditions, and sports participation patterns—should be carefully considered when interpreting normative benchmarks.

Overall, this study provides valuable population-specific reference data for Lebanese adults while contributing to the global literature on body composition and neuromuscular performance. These findings can support practitioners in health screening, individualized training design, and rehabilitation programming, while also facilitating meaningful international comparisons.

CONCLUSION

This study is the first to establish normative reference values for handgrip strength, explosive strength, and body composition in Lebanese adults aged 18–40 years. Resistance training was associated with favorable adaptations, including greater skeletal muscle mass, fat-free mass, and grip strength, along with reduced body fat. Women demonstrated more pronounced relative gains in maximal and explosive strength than men, underscoring sex-specific adaptive patterns. These findings provide a practical framework for evaluating performance, prescribing training programs, and monitoring rehabilitation in Lebanese adults.

The cross-sectional design and reliance on a convenience sample limit causal inference and generalizability. Future research should adopt longitudinal approaches, examine different training modalities, and extend normative profiling to wider age groups and elite athletes. Overall, this work contributes foundational benchmarks, advances understanding of sex-specific adaptations, and supports evidence-based strategies for performance evaluation, training optimization, and health screening.

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DATA AVAILABILITY STATEMENT

The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request. Individual participant data are anonymized and stored securely in accordance with institutional guidelines.

AUTHOR CONTRIBUTION STATEMENT

Zahi Gergi Andraos is the sole author of this manuscript and was exclusively responsible for all components of the research process. His contributions include the conceptualization and design of the study, data collection and management, execution of statistical analyses, and interpretation of the findings. He also carried out the writing, revision, and final approval of the manuscript, while ensuring the accuracy, integrity, and scientific rigor of every aspect of the work.

ETHICAL APPROVAL

This study was conducted in accordance with the Declaration of Helsinki. Ethical approval was obtained from the Ethics Committee of the University of Belgrade, Faculty of Sport and Physical Education (Approval Number: 02 No. 484-2, Approval Date: 15 September 2023).

All participants provided informed consent prior to participation, and all procedures were performed following institutional ethical standards.

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